Received: March 05, 2023 | Accepted : March 30, 2023 | Published : April 06, 2023

ORIGINAL RESEARCH

Peercite
Journal of Agriculture and Food Research

THE APPLICATION OF BIOTECHNOLOGY IN ENHANCING FOOD COMPLEXITY

Charis M. Galanakis

Department of Food Waste Recovery Group, ISEKI Food Association, Vienna, Austria

★*Corresponding Author : Charis M. Galanakis*

Email:(cgalanakis@chemlab.gr)

DOI: http://doi.org/10.61641/pjafr.2023.1.2.6508-6513

Citation: Galanakis CM (2023) The application of biotechnology in enhancing food complexity. Peercite J Agric Food Res. 1(2):6508-6513.

Copyright: © 2023 by the authors. Licensee MDPI, Basel, Switzerland. This article is an open access article distributed under the terms and conditions of the Creative Commons Attribution (CC BY) license (https:// peercite.org).

Abstract :

While the sustainability of the food enterprise is predicated on attempted-and-genuine technologies, there are instances in which these technologies end up outdated. thus, a key aspect for achievement in food businesses is the capacity to constantly innovate, adapt to the ever-changing panorama, and keep a competitive facet within the marketplace. At present, one of the most disruptive manufacturing technology in the meals region is 3D printing, that is drawing close an international marketplace well worth nearly a billion bucks and has skilled vast annual boom. This generation can revolutionize meals texture design and generate tailor-made products with particular traits, flavors, hues, textures, and nutritional profiles. In addition, it empowers the customization of food based totally on man or woman choices and desires, with packages ranging from area meals and restaurant cuisine to exceptional 3D edibles, such as jellies, dough, pizza, pasta, biscuits and candies.

Keywords : Food waste, Protein, Biotechnology

Introduction :

It includes multi-fabric printing and complex inner systems, allowing the improvement of healthier food products with reduced sugar, salt, and oil contents. It addresses approaching food and environmental challenges, which includes integrating dried wheat worms into diets and diversifying flour. However, further studies are wanted to optimize printing substances and parameters and to recognize how fabric rheological residences effect approaches consisting of extrusion. Addressing the demanding situations related to consumer attitudes, cognizance, standardization, and meals material consistency regarding 3-D-revealed foods is crucial [1].

Meals Processing and food era:

Driven by means of the complexities of modern-day deliver chains and client options for secure meals, clever packaging (which incorporates energetic and sensible technologies) is any other innovation that will become increasingly sizeable as it enhances the safety and great of meals products. Lively packaging consists of elements aimed at keeping or extending the shelf lifestyles of merchandise. On the identical time, shrewd systems reveal packaged food conditions (e.g., temperature and pH) all through storage and transportation without releasing any components into the food because of strict policies, Europe is currently behind in adopting lively and smart packaging [2].

However, the future meals enterprise can be directed closer to this direction for the reason that smart packaging encompasses numerous advantages in protection, advertising and marketing, and logistics.

Similarly, the call for safer and greater customized meals merchandise and the current sustainability demanding situations have pressured the meals industry to lessen strength and useful resource intake. Integrating non-thermal technologies into meals processing is becoming increasingly vital to maximize the conversion of raw materials into purchaser merchandise. Moreover, conventional processing and food separation techniques, like solvent extraction and membrane filtration, regularly lessen yields, decrease bioactive compounds' content material, and decline food fine [3]. Non-thermal technologies offer gentler processing situations and food treatment without destroying their sensory and nutritional characteristics attributes while retaining the functionality of bioactive components throughout extended garage. These technologies include radio frequency drying, excessive-voltage electrical discharge, high-strain processing, pulsed electric powered field processing, and microwave-assisted, ultrasound-assisted, and supercritical fluid extractions. Further studies within the forthcoming. Years can cope with the ability boundaries of those technologies, which includes their effects on bioactive lipids and interactions amongst meals bio-actives. It's also necessary to lessen carbon footprints by adopting power-efficient appliances, renewable electricity assets, sustainable products, transportation practices, and agricultural methods [4].

Impact of Biotechnology:

Crop development is crucial for meeting the world's growing food, feed, and bioenergy needs. It has advanced from traditional breeding strategies (e.g., hybridization and selection) to proteomics and revolutionized biotechnological tools, integrating genetic engineering, genomics, and precision breeding that enhances productivity and flexibility to converting environments [3,5]. As an example, environmental elements, which include drought, salinity, and biotic stresses, in particular associated with climate exchange set off demanding situations to crop growth. Proteomics presents insights into plant molecular mechanisms, permitting the improvement of stress-resistant, high-yield crops through marker-assisted breeding through unraveling molecular mechanisms and revealing tricky proteomic conduct related to plant strain tolerance. Moreover, via introducing focused genes or employing superior gene-modifying equipment, researchers can very well alter crop characteristics, boosting them with resistance to pests, sicknesses, and environmental stresses [1,6].

To this end, the lustered, regularly interspaced brief palindromic repeat (CRISPR)/CRISPRassociated protein (Cas) is an effective genome editing tool that can be successfully used for particular genome enhancing of wheat and other crops. This system has revolutionized crop improvement with its simplicity and multiplexed gene editing skills, permitting speedy improvements in less than a 12 months in comparison to traditional methods that take 6-7 years. Its applications in disease resistance, drought tolerance, and pleasant enhancement show promise for addressing food needs inside the face of a developing population. Further, integrating CRISPR/Cas9 with next-generation sequencing and advanced genomic methods enhances mutational screening and useful genomics, in addition propelling crop development. Latest CRISPR/Cas applications in plants like Arabidopsis, rice, maize, soybean, and tobacco have stepped forward yield and strain tolerance developments while heading off stringent regulations and moral worries associated with permanent genetic modification [3,7].

The challenges encompass designing correct guide RNAs, selecting promoters, overcoming size obstacles for viral packaging, and creating green, tissue way of life-unfastened modifying techniques. CRISPR technology is likewise an emerging tool for ultrasensitive pathogen and contaminant detection in packaged meals, supplying actual-time surveillance and improving patron health safety. However, at the same time as genetically modified crops have the potential to enhance yields, in addition they increase concerns regarding biosafety and environmental worries, like altered crop conduct, herbicide and insecticide tolerance levels, transgene stacking, and biodiversity disruption. Ongoing studies and knowledgeable choice making are crucial for assessing their influences and guiding their cultivation and use [2].

Cultural range, Culinary developments, and Sustainable meals intake:

Globalization has fostered an international meals lifestyle, transcended borders, and promoted cultural appreciation via meals. However, following a few food scandals and conflicts associated with genetically changed meals manufacturing which have undermined public accept as true with, purchasers within the present day era are displaying a growing desire for traditional, domestically sourced, natural, and slow meals.

The latter fashion is expected to preserve focusing on traditional, domestically sourced, and sustainable food practices, appealing to customers in search of an extra proper and ethical eating enjoy. Though, traditional meals producers are continuously challenged to decorate the protection and comfort in their products to fulfill market needs at the same time as additionally grappling with the strain precipitated through large shops, open markets, and compliance with rules in the end, they extend their abilities and undertake improvements to spotlight their products' nutritional benefits and health blessings but, converting methods dangers changing the conventional appeal, that is met with consumer resistance, in particular inside the traditional food region [4].

Driven with the aid of cultural variety, culinary innovation plays a critical function on this direction, encompassing the exploration of unique flavors and adapting seasonal variations and local elements. In addition, it allows the fusion of culinary traditions, emphasizing neighborhood factor sourcing, which now not only preserves cultural traditions, but also aligns with sustainability goals, decreasing carbon footprints and supporting neighborhood communities. Despite the fact that, we need to keep away from assuming that neighborhood food routinely has a more negligible environmental impact than worldwide food clearly due to the fact it's miles produced toward consumption. For example, eating places are ordinary resources of environmental unsustainability that may occur both from wastages (food and non-biodegradable materials like plastics) and emissions (burning of fuels inside the kitchen and vicarious resources like food sourcing and deliver) [2].

Instead of simplifying food into "nearby" vs. "international" or "in season," it's far really helpful to undertake a holistic attitude of the meals system, recognizing the interplay among local and international factors among food actors, fostering methodological approaches addressing sustainability across all dimensions and facilitating tangible development in sustainable food intake. To this quit, traditional and indigenous food systems are inherently integrating sustainable practices, which includes agroforestry and coffee-impact fishing. Furthermore, inexperienced and sustainable dining practices, which include farm-to-desk concepts, zero-waste initiatives, and environmentally conscious menu choices, are continuously gaining recognition, enjoyable patron choices and environmental goals. Sustainability practices in meal manufacturing, practise, and carrier are critical to fashionable diners' notion of "overall exceptional." This trend includes services enjoyable customers' nutrients, hygiene, social, and cultural requirements. Foodservice businesses expected to develop consist of the ones specializing in meals shipping and takeout offerings, driven by way of the increasing demand for comfort and the increase of on line ordering structures [8].

Additionally, health-focused and plant-primarily based foodservice carriers are probable to look growth as customers prioritize healthier and sustainable eating options. Possibilities also exist for nearby governments to implement policies that promote each human and planetary health, including measures to reduce food overconsumption, such as encouraging lower meat intake and regulating ultra-processed meals. Inclusivity in dining, addressing numerous nutritional wishes and cultural backgrounds, promotes an experience of belonging [7]. Moreover, technological disruptions form the destiny of eating experiences. As an example, a proposed innovation includes a clever dining table that measures food weight to tune consumed energy. Fast meals will probably emphasize healthier menu options, sustainability, and more suitable generation for ordering and transport. Augmented and virtual truth revolutionary technologies offer immersive culinary adventures, influencing purchasers' behavioral, sensory, social, and highbrow perceptions.

Therefore, restaurants can use augmented fact to create immersive reviews (e.g., thru aesthetics and storytelling) that boost the overall meals nicely-being in their clients, integrating their choices. Moreover, there are guidelines for a machine that makes use of cameras to analyze food pictures, identify meal additives and portion sizes, and estimate calorie intake via on-line databases [9].

Personalised vitamins:

The whole lot we devour is broken down into molecules that uniquely engage with our microbiome and genome. For example, probiotics can help with irritable bowel syndrome and prevent colorectal most cancers. At the identical time, nutrients A and C possess anti-carcinogenic houses, and folic acid prevents DNA mutations with the aid of enhancing protein methylation strategies.

Ultimately, fitness can depend on diets and micronutrients to reduce genetic mutations causing diseases. Inside the present nutritional panorama, the traditional belief of "one-size-fits-all" dietary recommendations have given way to a more modern technique driven via individuals' particular alternatives [2]. This modification has brought about the emergence of personalized nutrition, recognizing that absolutely everyone is distinct and responds in a different way to the identical meals. Customized vitamins follow the development of the nutrigenomics subject that research the relationship between nutrition and gene expression.

. Nutrigenomics explores greater powerful and customized strategies to weight-reduction plan primarily based on the study of man or woman genetic and physiological variations influencing nutrient tiers and dietary reactions, imparting promising avenues for cancer prevention and the remedy of illnesses [3].

Personalised vitamins integrates an individual's dietary options and sustainability values into dietary tips, guiding them in the direction of food choices. In an international regularly facing weight problems and related metabolic issues including kind 2 diabetes and cardiovascular illnesses, personalised dietary pointers offer promise for prevention strategies. Challenges include the complex polygenic nature of nutrition related illnesses, necessitating the consideration of a couple of genetic versions in predictive models and knowledge gene-environment interactions, especially in sicknesses motivated by way of outside elements. Reaching this intention necessitates a collaboration between fundamental and scientific scientists and healthcare experts to establish a comprehensive framework for implementing those discoveries on a populace scale [10].

For meals companies, customized nutrients represent an enormous road for innovation, permitting them to create tailor-made products and nutritional solutions that align with consumers' needs. Nutrigenomics and deep phenotyping are just a few elements that should be taken into consideration to craft personalised and independent dietary solutions. Advanced omics technologies can also offer insights on this direction, however they require gadget getting to know and different digital gear to improve fitness outcomes and decrease ailment burdens. As an instance, imposing system gaining knowledge of and synthetic intelligence for predicting weight loss and fitness outcomes (e.g., thru random woodland fashions and neural networks) has the ability to enable the identity of factors affecting the prediction of a man or woman's weight reduction progress. This sort of version has the ability to assist and supplement healthcare experts in recommending tailored changes to electricity consumption and growing personalized nutrients plans. Alongside this line, the ability of wearable and cell chemical sensors to tune and manual nutrients is predicted to develop quite over time to come. Moreover, numerous technology, like microphones, piezoelectric sensors, and accelerometers, have been demonstrated to reveal moves such as biting, chewing, swallowing, and arm/wrist movements to assess calorie intake based on bodily interest. Integrating various sensor modalities (e.g., non-invasive wearable electrochemical ones) with data analysis strategies can bridge the space between digital and biochemical analyses, and precision vitamins may be revolutionized, making customized vitamins more effective [2,6].

Patron fitness and health:

The superiority of chronic illnesses as a leading worldwide purpose of dying, associated with the overconsumption of processed meals and inadequate consumption of plant-based ingredients, has brought about a rise in fitness-aware people adopting nicely-being techniques (e.g., the Mediterranean weight loss plan and expanded physical activity) but, these strategies can be complicated and impractical for many in modern-day societies. Consequently, many customers have redirected their interest toward the energetic position of natural supplements. Over the past a long time, there was a widespread clinical cognizance on figuring out the goal active components in plant-primarily based ingredients and wholesome diets, exploring their potential anti-oxidative, anti-tumor, anti-hyperlipidemic, anti-hypertensive, antimicrobial, and anti-viral sports, further to their essential dietary features [7].

Along this line, fortifying ingredients and boosting diets with nutritional additives fostered innovation within the food enterprise surroundings, main to the substantial use of phrases which include nutraceuticals. Even though missing a universally prevalent definition, nutraceuticals are medicinal retailers owning drug-like attributes able to successfully addressing excessive ailments, like diabetes, atherosclerosis, most cancers, neurodegenerative disorders, and hematological conditions [2]. They harness the capability of bioactive compounds, which includes tannins, polyphenols, flavonoids, alkaloids, and terpenoids, normally determined in wholesome meals products. Alternatively, useful meals are generally recognized as products with demonstrated blessings in improving precise physiological capabilities beyond fundamental nutrients, contributing to more advantageous fitness and reduced disorder risks [8]. Useful and custom-designed meals are created by incorporating precise substances into traditional or conventional meals merchandise and altering their residences through binding, structural adjustments, or interface modifications, all while imparting fitness-improving traits to these merchandise [1].

Food bio actives and nutraceuticals stumble upon demanding situations associated with their intricate compositions and diverse modes of movement, encompassing obligations like extraction, chemical characterization, assessing in vitro and in vivo bioactivity, and comprehending their interactions with organs and microbiota.

. Their effectiveness relies upon on multiple factors, encompassing bioactivity, bioavailability, metabolomics, nutrigenomics, and their stability within the food matrix. For example, when administered orally, bio actives face constraints imposed via gastrointestinal enzymes, acidic pH stages, the epithelial barrier, and the mucosal layer. Not with standing worries approximately efficacy, great, medical evidence, and self-remedy risks, the demand for nutraceuticals and useful ingredients keeps to develop, creating opportunities in the marketplace, that is expected to flourish within the submit-lockdown length, driven by means of the improved attention from fitness-aware purchasers. Nevertheless, growing novel practical foods is multifaceted and financially stressful, with patron adoption contingent upon severa influencing elements [8]. Comprehending consumer uncertainty and skepticism regarding those products is critical for effective product development, and a deep understanding of customer choices and attitudes is critical for agencies to adapt. Practical ingredients allow purchasers to improve their health whilst taking part in familiar and handy nutritional alternatives. Nonetheless, it's miles essential to make sure that the claims of health benefits are based totally on sound medical proof, and protection concerns have to always be a concern [11].

Meals manufacturing inside the round Bio economy:

Food bio actives can be sourced from with the aid of-merchandise generated for the duration of food processing. These materials are abundant in useful compounds like antioxidants (polyphenols and carotenoids), glucosinolates, proteins, and nutritional fiber (pectin and β-glucan). On the equal time, they can be re-used to support meals merchandise and cosmetics. Moreover, the crucial role of meals waste valorisations is pivotal within the rising technology, forming the inspiration of the bio economy, which, in flip, constitutes the core of the circular financial system [10].

This circular approach aligns with the dreams of weather-impartial economies aimed for by means of advanced nations like Europe, the us, Japan, and China by way of 2050-2060. Likewise, making use of inedible and unavoidable meals waste and residues as bio refinery feedstock aligns with United Countries Sustainable improvement goal 12.3, focused on a reduction in meals losses across the manufacturing and supply chain [9].

The transition of the cutting-edge linear development model to a round bio-economy method can enhance resilience by way of offering opportunities to valorises food waste and different biomass substrates.

Accomplishing this purpose includes addressing technical, financial, and clinical challenges through a multidisciplinary method enabled by way of the bio refinery idea, which integrates approaches to convert resources into various bio-primarily based merchandise. Along those lines, pyrolysis and hydrothermal carbonization can convert biomass successfully into environmentally friendly biofuels, bio-chemicals, and precious carbon-based substances [4]. Bioenergy is a treasured product that may be applied for energy generation, heat production, and transportation, offering a sustainable answer that mitigates landfill-related troubles and GHG gas emissions. On the other hand, at the same time as meals waste valorisation is a promising discipline, posing demanding situations as an ordinary solution does no longer exist because of the varied nature of meals waste and the need to confirm feedstock composition and desired very last products [2]. The green valorization of meals waste inside the bioeconomy calls for addressing feedstock variability, era choice, improving product performance, decreasing manufacturing expenses, and innovating separation techniques. Moreover, the primary purpose continues to be to reduce meals waste, and healing tactics need to healthy nearby conditions without disrupting the meals supply or geopolitics. Efforts to beautify resource efficiency ought to account for the capability rebound impact, where greater efficiency can cause multiplied resource consumption. Future breakthroughs in meals waste valorization and control can contain the metabolic engineering of microorganisms for product specificity [7]. Sooner or later, an interdisciplinary collaboration is important for cost-powerful hybrid conversion systems, with manner models predicting the monetary cost to draw investments. However, scaling needs comprehensive product restoration and environmental effect checks [3].

Destiny perspectives:

Attaining sustainable food manufacturing in the decades to return is a multifaceted project that requires comprehensive modifications across diverse aspects of the food supply chain. Embracing crop rotations, natural farming, agroforestry, and vertical farming is essential for optimizing agricultural productiveness and aid control. Moreover, the destiny of food is based on modern procedures that balance vitamins, fitness, and environmental responsibility with innovations which includes three-D printing, smart packaging, and non-thermal technologies, food processing technologies hold to conform [2,9].

The digitalization of the food deliver chain thru enterprise 4.0 technology offers incredible capacity to deal with the challenges and decorate traceability, transparency, and meals safety even as optimizing supply chains, reducing meals loss and waste, and promoting meals safety on the identical time. Food security and climate-resilient supply chains are certain to alternative protein resources and the usage of meals processing through-products and waste that align with a climate neutral, round bio-economy and sustainable development desires.

Crop productivity and food security may be more suitable by using genome enhancing equipment (e.g., CRISPR/Cas9) [2,9]. The ongoing studies addresses the demanding situations related to nutritional completeness, client acceptance, protection, and policies. Ultimately, the evolving food landscape embraces cultural variety, culinary innovation, and sustainability developments, emphasizing traditional, locally sourced, and organic meals, sustainable eating practices, and personalized vitamins. On this ever-evolving food landscape, our collective commitment to innovation, sustainability, and the properly-being of each human beings and the planet will shape a destiny where food inspires a superb alternate and resilience for future generations [5].

Conflict of interest:

Author declares no conflict of interest.

References:

1.Ghoulem M, Moueddeb KE, Nehdi, E et al. (2019) Greenhouse design and cooling technologies for sustainable food cultivation in hot climates: Review of current practice and future status. Biosyst Eng. 183:121-150.

2.Rodríguez MP, Domínguez A, Moreira Ferroni M et al. (2020) The diversification and intensification of crop rotations under no-till promote earthworm abundance and biomass. Agronomy. 10:919.

3.Jose S (2009) Agroforestry for ecosystem services and environmental benefits: An overview. Agrofor Syst. 76:1-10.

4.Mosquera-Losada MR, Santiago-Freijanes JJ, Rois-Díaz M et al. (2018) Agroforestry in Europe: A land management policy tool to combat climate change. Land Use Policy. 78:603-613.

5.Skar SLG, Pineda-Martos R, Timpe A et al. (2019) Urban agriculture as a keystone contribution towards securing sustainable and healthy development for cities in the future. Blue-Green Syst. 2:1-27.

6.Omar S (2021) Internet of Things (IoT) for Smart Farming: A Systematic Review. Int J Comput Appl 174:47-54.

7.Abu NS, Bukhari WM, Ong CH et al. (2022) Internet of Things Applications in Precision Agriculture: A Review. J Robot Control. 3:2715-5072.

8.Rejeb A, Rejeb K, Abdollahi A et al. (2022) Digitalization in Food Supply Chains: A Bibliometric Review and Key-Route Main Path Analysis. Sustainability. 14:83.

9.Sempionatto JR, Montiel VRV, Vargas E et al. (2021) Wearable and mobile sensors for personalized nutrition. ACS Sens. 6:1745-1760.

10.Batat W (2021) How Augmented Reality (AR) Is Transforming the Restaurant Sector: Investigating the Impact of 'Le Petit Chef' on Customers' Dining Experiences. Technol Forecast Soc Chang. 172:121013.

11.Alshurafa N, Kalantarian H, Pourhomayoun M et al. (2015) Recognition of Nutrition Intake Using Time-Frequency Decomposition in a Wearable Necklace Using a Piezoelectric Sensor. IEEE Sens J. 15:3909-3916.